1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

TLS-like lymphoid aggregate-rich tumors with high HRD scores and elevated MDK exosomes fail to benefit from therapy because tumor-derived exosomal MDK reprograms the local immune niche, impairing TLS maturation and antigen presentation despite a high neoantigen load. Clinically, HRD-high/TLS-high tumors with elevated MDK will show reduced therapy_response (to DNA-damaging agents and/or immunotherapy) compared with HRD-high/TLS-high tumors with low MDK, identifying MDK as a targetable modifier to improve therapeutic sensitivity.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	HRD_score, MDK
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

HRD_score: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

MDK: MDK/KRT17 path (core path 3) as a mechanism for tumor-driven conditioning of the microenvironment.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	KRT17	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph

Secondary	BRCA1	Supporting modifier from knowledge graph
Secondary	MSH6	Supporting modifier from knowledge graph
Secondary	PARP1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: and MDK in tumor sections; quantify exosomal MDK in patient plasma by ELISA or targeted proteomics; correlate with response. - In vitro mechanistic studies: generate tumor-derived exosomes with/without MDK (knockdown/overexpression) and assess effects on dendritic cell maturation, antigen presentation (MHC II, co-stimulatory molecules), and T/B cell activation in TLS-model cultures.

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- KRT17: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- HDL-C: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.